

10 Years of RStudio

And What Happens Next

A Play in 3 Acts

Act 1: RStudio

Act 2: Positron

Act 3: What Happens Next

Act 1:

RStudio

2013





SELF SERVE

CAR WASH

SELF SERVE

REGULAR
LEADED

83⁹/₁₀

REGULAR
UNLEADED

88⁹/₁₀

EXXON

Shop

Self ⚡ Cash

Exxon
Regular 83⁹/₁₀

Exxon
Unleaded 87⁹/₁₀

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Self-Service
Supplies
1021

Welcome to RStudio

Software, education, and services for
the R community



Powerful IDE for R

RStudio IDE is a powerful and productive user interface for R. It's free and open source, and works great on Windows, Mac, and Linux.

[Download now](#)[Learn more](#)

R training and education

We've got hands-on courses for beginners and even R experts. Customize an on-site training or enroll in one of our public workshops.

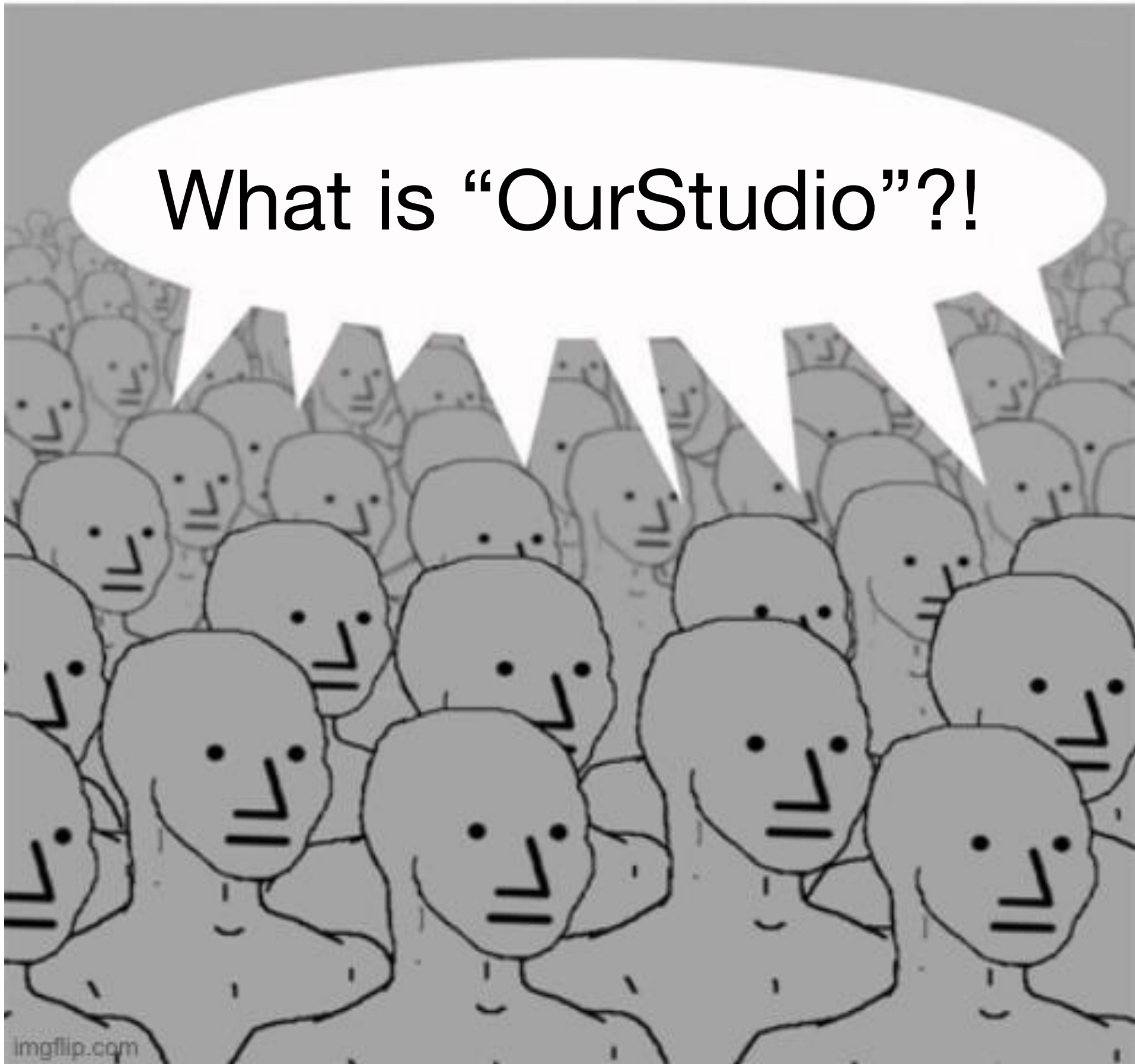
[Request on site](#)[View courses](#)

Open source R packages

Our developers and expert trainers are the authors of several popular R packages, including ggplot2, plyr, lubridate, and others.

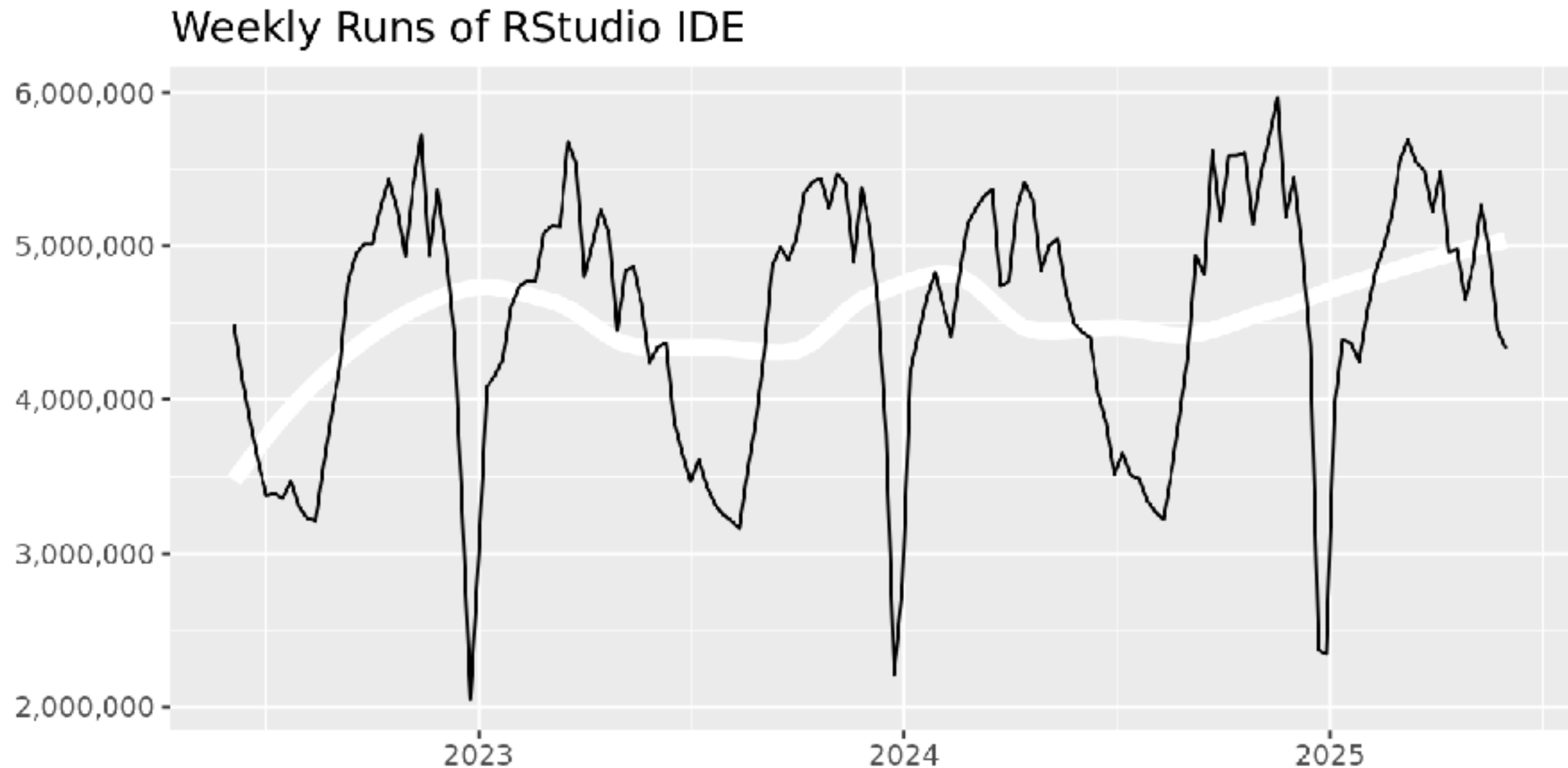
[See projects](#)

What is “OurStudio”?!



Today (June 2025)

Weekly use 5M and growing



**Why has RStudio been
so successful?**

Made by an R user

(Not a focus group)



Commit **c719a59**

 **jjallaire** committed on Dec 7, 2010

`first commit`

 **main** ·  v2025.05.1+513 ... v0.92.23

Free & Open

- Available for every platform & the web
- Permissive AGPL license
- Posit is a Public Benefit Corporation
- We can (and do) build for our mission over profit
- No VC driven dark patterns or revenue grabs



**open source
initiative®**

Build for a single purpose

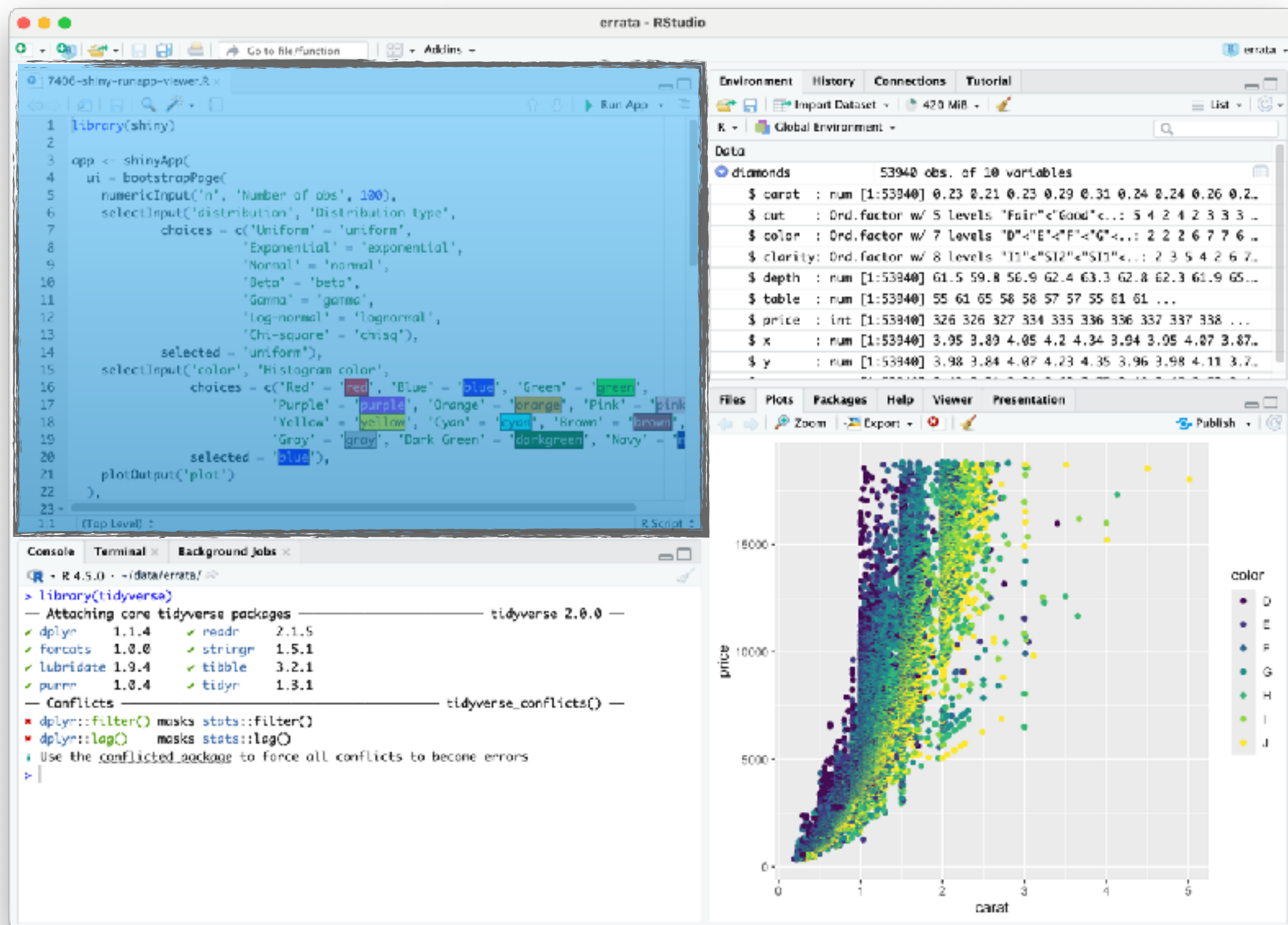
A specific tool beats a general-purpose tool



It's okay	It's okay
Utter bliss	Possible but very messy Why would you do this
No	Deep satisfaction

Work with data and results

(Not just code!)



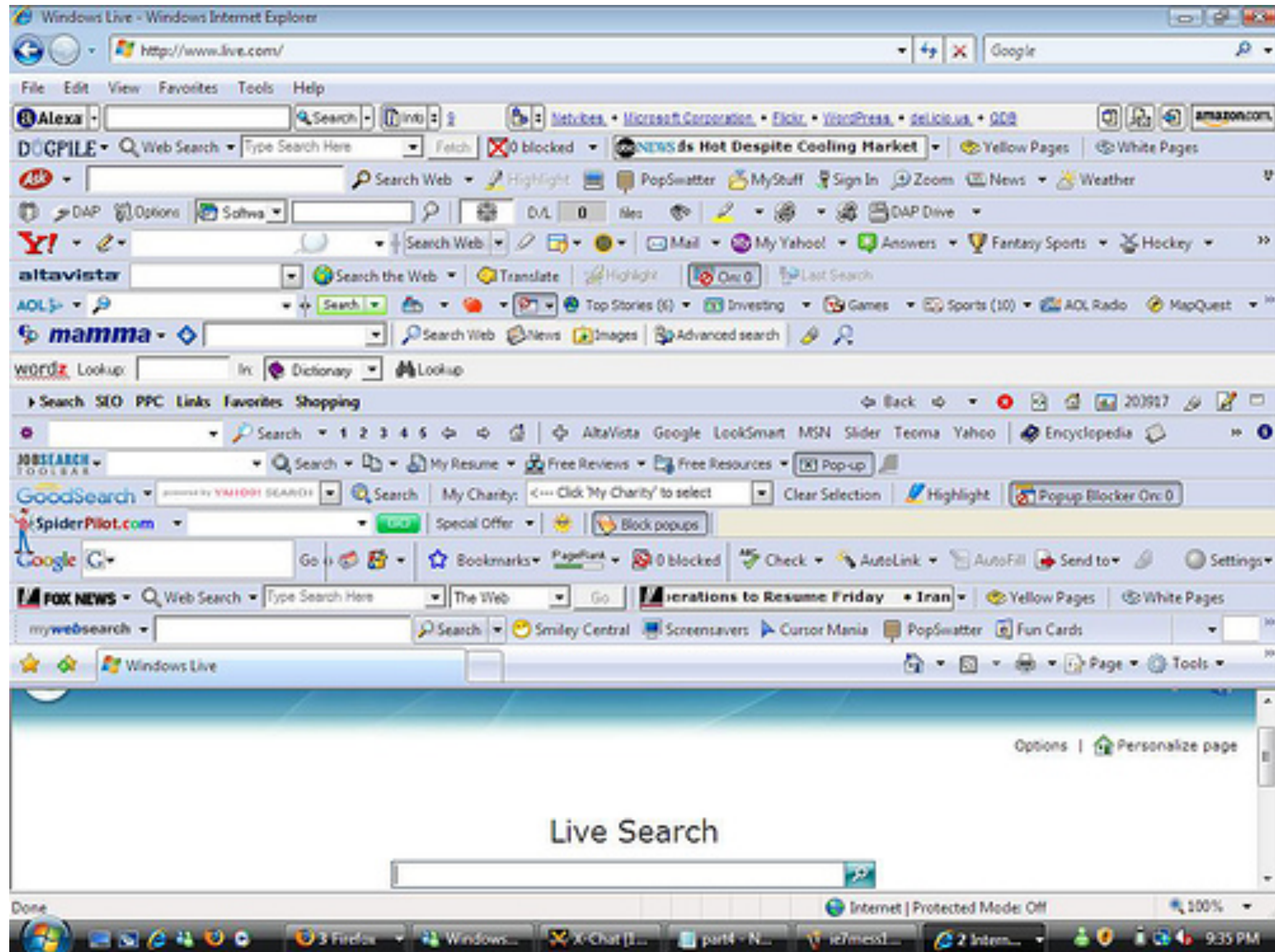
Starts ready to use

If R is installed, it works.

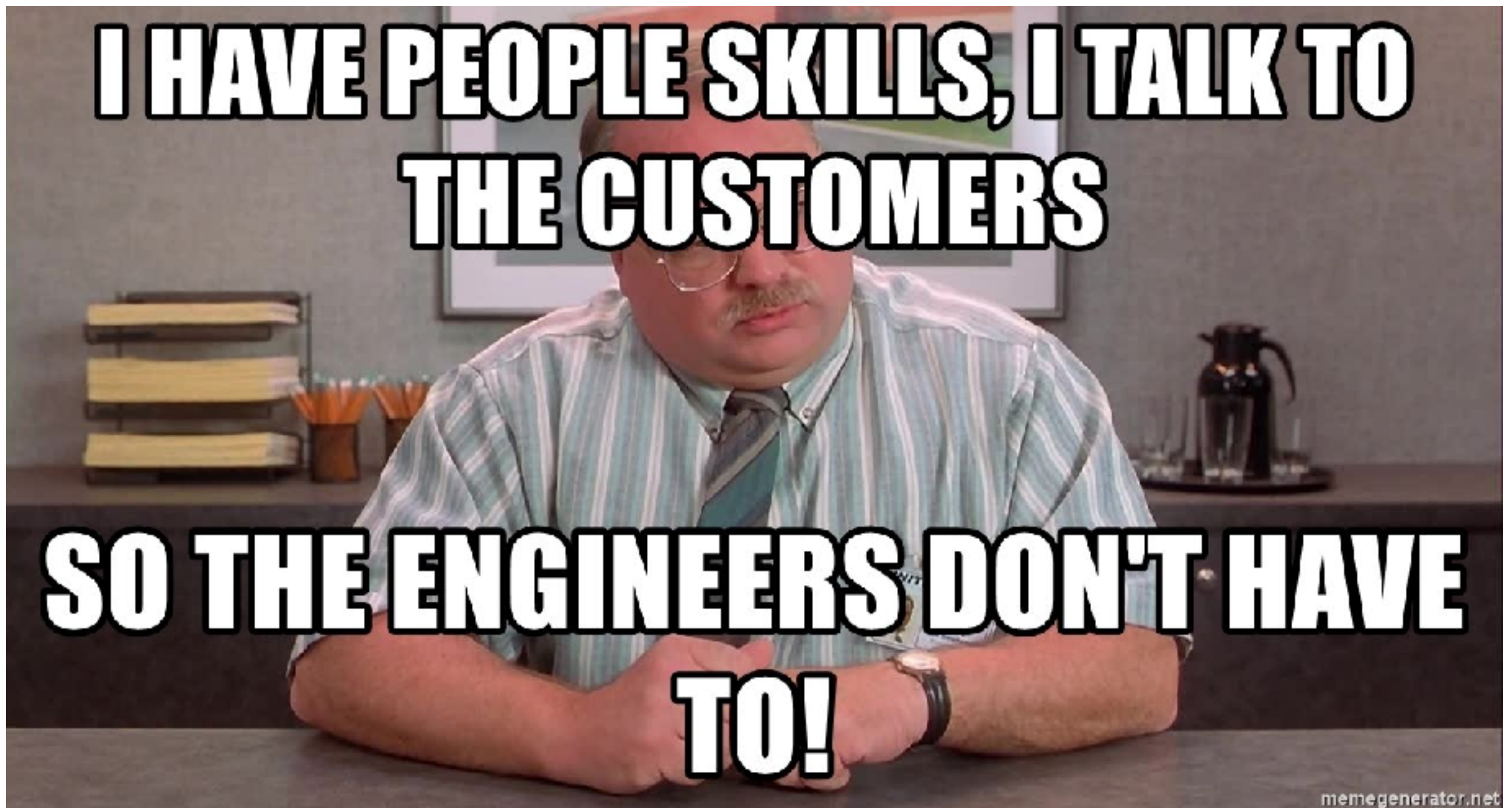


The image shows a screenshot of an R console window. The window has a title bar with three tabs: 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. Below the tabs, there is a status bar that reads 'R 4.5.0 · ~/data/errata/'. The main area of the console displays the following text: '> 2 + 2', '[1] 4', and '> |'. The prompt character '>' is blue, and the output '[1] 4' is black. The cursor is positioned at the end of the prompt '> |'.

Convention over configuration



Engineers & customers



Listen to your users

Feature request: format R codes on save #2563

✓ Closed



yutannihilation opened on Apr 1, 2018

Currently, RStudio can automatically format codes on save a bit:

- Ensure the source files end with newline
- Strip trailing horizontal whitespace when saving

As [styler](#) reached 1.0, how about integrating the feature with RStudio?

Or, is it possible to let users to insert "before save hook"?

In my understanding, "format on save" is a very common feature in modern editors/IDEs. For example, VS Code lets users to choose the formatter and when/whether to format the code or not:

https://code.visualstudio.com/docs/editor/codebasics#_formatting

For another example, Emacs doesn't have direct "format on save", but users can define

<https://github.com/dominikh/go-mode.el/blob/ff87a1e80945d958d9c348ce4d2e65a79>

I also believe "format on save" is useful to soften the frustration from the incompleteness

Create sub-issue



👍 74



👍 74

Empower the community

Discover and install useful RStudio addins

 R-CMD-check passing  CRAN 0.5.0  Donate  PayPal

Created by [Dean Attali](#)

[RStudio addins](#) were released in early 2016 to provide anyone with the ability to add "extensions" to RStudio. This feature has quickly become popular, but discoverability was a problem: **there's just no easy way to know what addins exist.**

The `addinslist` package solves that problem in two ways:

- This package provides a continuously updated list of RStudio addins that you can browse through ([below](#)).

To sum up

- Start small, start organically
- Don't try to make something for everyone (especially not at first)
- Develop freely and in the open
- Connect engineers directly with users
- Make it simple to start and use
- Empower, love, and listen to your community

Act 2: Positron



**If RStudio is so great, why
did you make Positron?!**

RStudio's limitations

- Single-session
- Single-language
- Limited IDE features
- Zero-sum contest between data science vs. IDE features

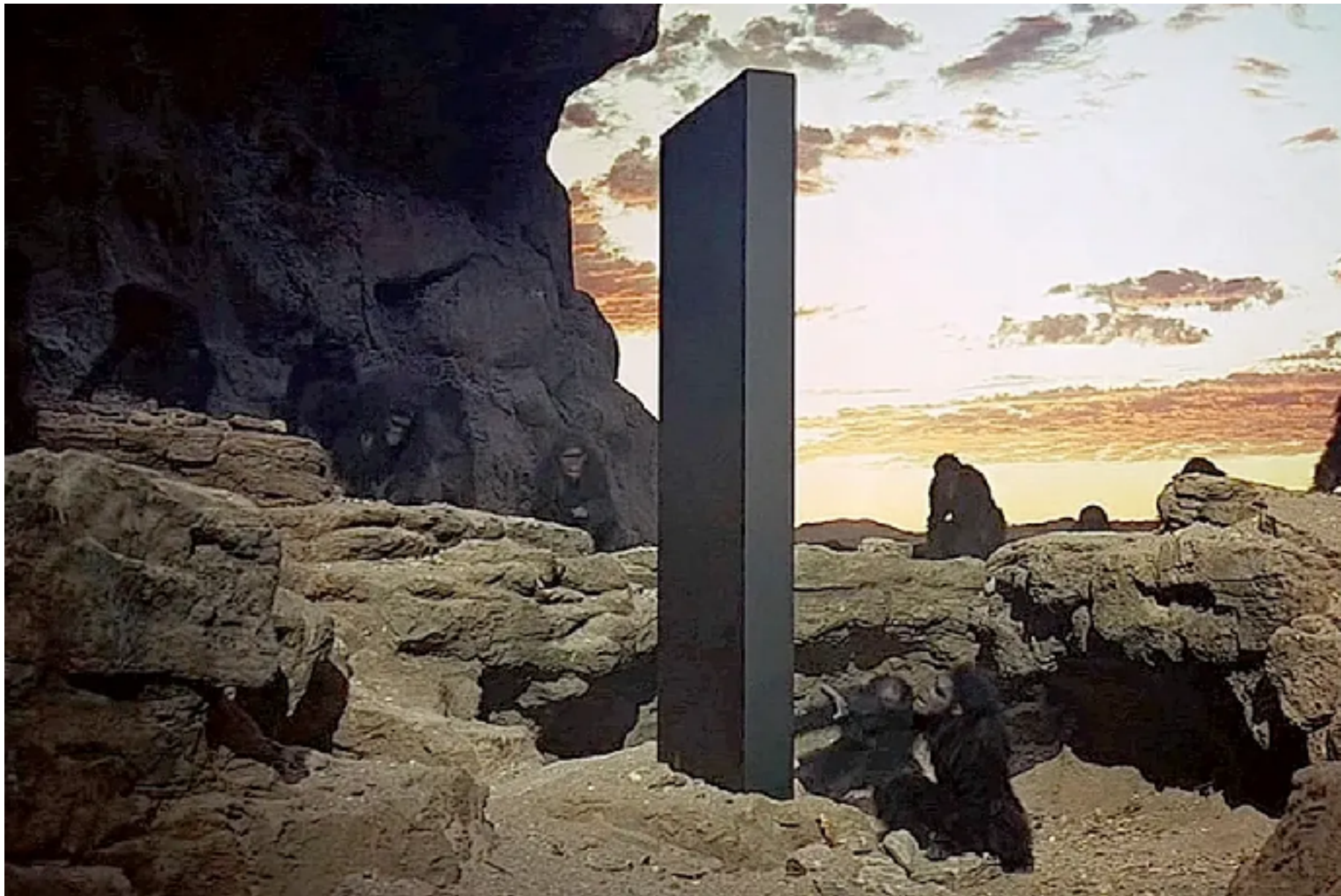
Positron's goals

- Multiple languages
 - Bring the best data science tools to *all* data scientists
- Multiple sessions
- Best in class IDE features
- Tool unification
 - Lots of people use one tool for data & VS Code for “everything else”

Act 3:

What Happens Next

Composable, open tools



Composable, open tools



R execution

R debugging

R code formatting

Data viewing

R code linting

Quarto authoring

R autocomplete

Variable viewing

Composable, open tools



Data viewing

R code formatting

R debugging

Variable viewing

R autocomplete

R execution

R code linting

tree-sitter-r on Github

The screenshot shows a GitHub search interface. At the top, a search bar contains the text "repo:rstudio/vetiver-r vetiver_model". Below the search bar, there are three search suggestions:

- Q repo:rstudio/vetiver-r vetiver_model Search in this repository
- Q org:rstudio vetiver_model Search in this organization
- Q vetiver_model Search all of GitHub

Below the suggestions, the "Code" section is displayed, showing a list of files with their relative paths and a "Jump to" link for each:

- `function vetiver_model` (rstudio/vetiver-r · R/vetiver-model.R) Jump to
- `skeleton.Rmd` (rstudio/vetiver-r · inst/rmarkdown/templates/vetiver_model_card/skeleton/) Jump to
- `template.yaml` (rstudio/vetiver-r · inst/rmarkdown/templates/vetiver_model_card/) Jump to
- `vetiver_model.Rd` (rstudio/vetiver-r · man/) Jump to

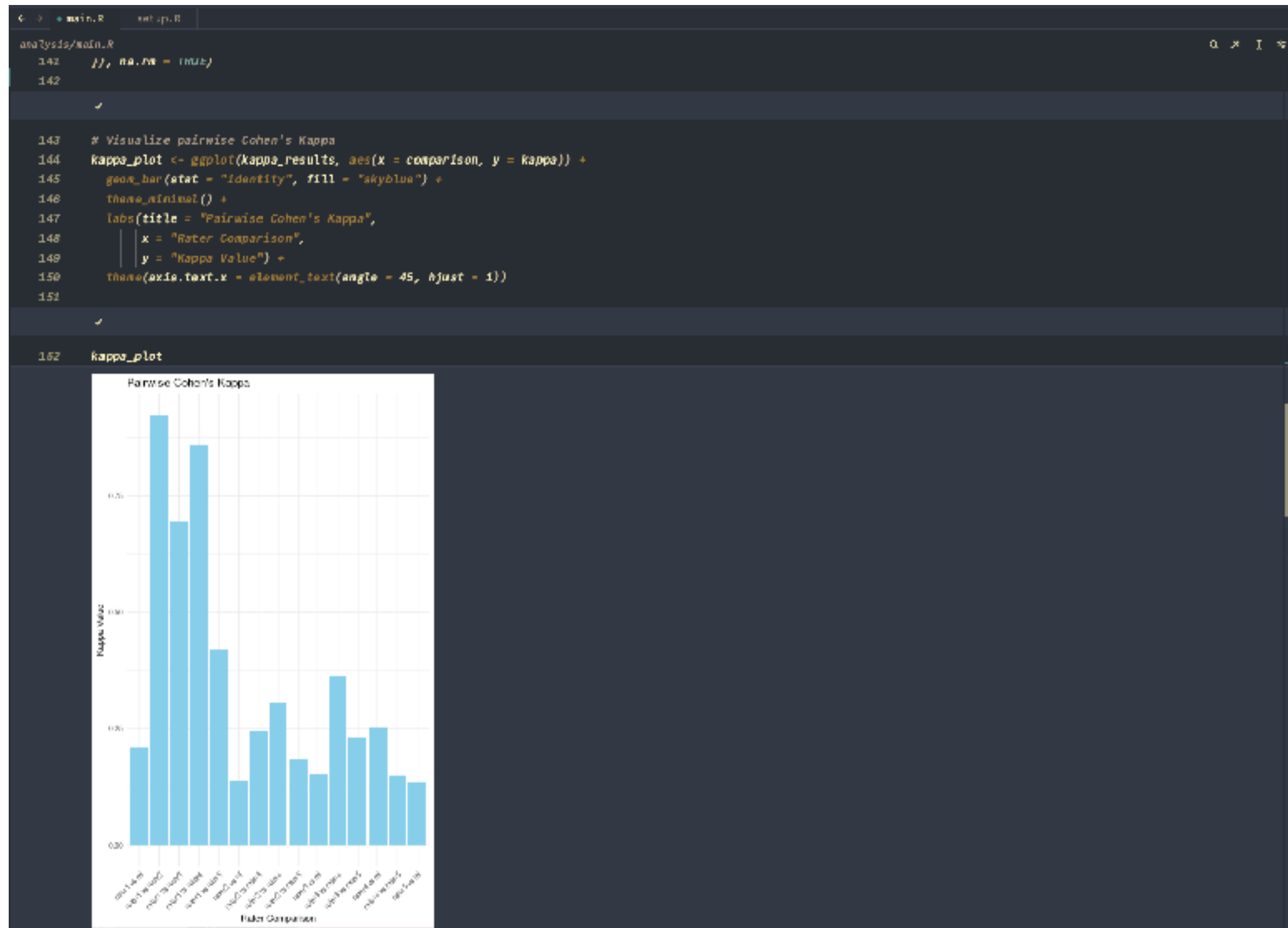
At the bottom of the search results, there are two links: "Search syntax tips" and "Give feedback".

Below the search interface, there is a snippet of R code:

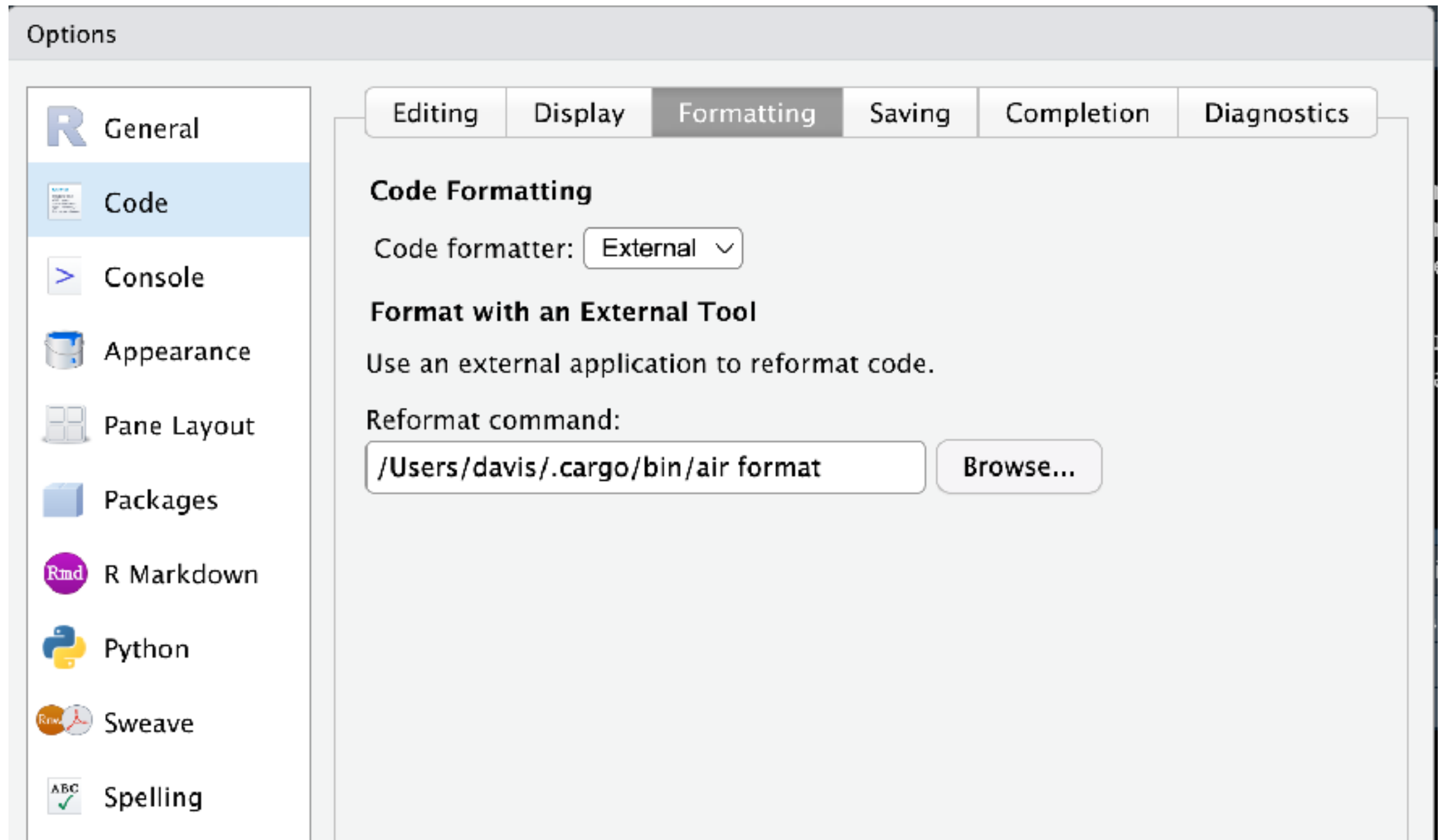
```
connect bundle for a vetiver model API
```

```
e_rsconnect_bundle()` to create a
```

Zed editor



Even RStudio!





positTM



What about RStudio?



Good!	Good!
Utter bliss	Do you like Reticulate?
No	Again, no

AI in your IDE

(Are you tired of hearing about this?)

- Models that can see and understand your code, *and* data, *and* computational environment, and pair with you to accomplish your goals
- Mind-reading auto-complete (sometimes)
- Vindication of code-first philosophy: Code is what models read and write
- Gives you superpowers, doesn't replace you

Q & A